

# Case Study: Mall Customer Segmentation for a Marketing Campaign

## Business Scenario

A shopping mall wants to improve its marketing strategy. Currently, it sends the same promotional offers to every customer, resulting in poor response rates and unnecessary marketing costs.

As a Data Analyst, your task is to segment customers into similar groups using **K-Means Clustering** so that the marketing team can create personalized campaigns.

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## Business Problem

The mall management wants answers to the following questions:

- Which customers spend the most?
- Which customers have high income but spend very little?
- Which customers are budget-conscious?
- How can we target each customer group differently?

Since there are **no predefined labels**, you decide to use **K-Means Clustering**.

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## Dataset

The company provides the following customer information:

- Customer ID
  - Age
  - Annual Income (₹ Lakhs)
  - Spending Score (1–100)
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## Your Tasks

### Task 1

Load the dataset into Pandas.

### Task 2

Perform basic data exploration.

- Check for missing values.
- View summary statistics.
- Visualize the distribution of income and spending score.

### Task 3

Select appropriate features for clustering.

**Hint:** Choose only the features that are useful for customer segmentation.

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### Task 4

Use the **Elbow Method** to determine the optimal value of **K**.

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### Task 5

Train a K-Means model using the optimal number of clusters.

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### Task 6

Assign each customer to a cluster.

Add a new column called:

Customer\_Segment

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### Task 7

Visualize the clusters using a scatter plot.

Clearly display:

- Different customer groups
  - Cluster centroids
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### Task 8

Analyze each cluster from a business perspective.

For each cluster, answer:

- Is this group high income?
  - Is this group high spender?
  - Should the business target them?
  - What type of offers should they receive?
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## Final Business Deliverable

Prepare a short report (1–2 pages) answering the following:

1. How many customer segments did you identify?
  2. Describe each customer segment.
  3. Which segment should receive premium membership offers?
  4. Which segment should receive discount coupons?
  5. Which segment is least valuable?
  6. If the marketing budget is limited, which segment would you prioritize and why?
  7. Recommend three actionable strategies for the marketing team based on your clustering results.
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## Expected Outcome

By the end of this case study, you should be able to:

- Apply K-Means to solve a real business problem.

- Choose the optimal number of clusters.
- Interpret clusters in business terms rather than just mathematically.
- Convert analytical findings into actionable business recommendations, similar to what a data analyst would present to management.